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UKUDLA Data Science Certificate

Data Management

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UKUDLA
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for Sustainable and Resilient Food
Systems and Applied Agricultural
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Reproducible analysis can reproduce a mistake

The workflow runs

- code is versioned;
- packages are restored;
- the container starts; and
- the report is rendered.

But the data may be misunderstood

- yield uses the wrong unit;
- missing values are read as zero;
- flags are ignored; or
- a revised source replaces an earlier file.

Data management makes the meaning, origin, quality, organization, and permitted use of data explicit.

Data move through a lifecycle



- Decisions at one stage affect what is possible at later stages.
- Documentation and provenance connect the stages.
- Validation is repeated when data, requirements, or uses change.

This session moves from **motivation**, through the essential **concepts**, to their responsible **application** in a data workflow.

What does one row represent?

Dataset grain

The grain states what a single observation represents.

For agricultural statistics, one row may represent one:

- area;
- year;
- item;
- element; and
- unit.

Role	Example
Identifier	Area Code, Item Code
Time	Year
Measure	Value
Meaning	Element
Context	Unit, Flag

A key is the smallest set of variables expected to identify one row uniquely.

Food-systems data have different structures

Tabular structures

- Cross-section: many units at one time
- Time series: one unit through time
- Panel: many units through time
- Hierarchical: units nested in larger units

Spatial structures

- Points: farms or weather stations
- Polygons: districts or countries
- Grids: climate or satellite cells
- Boundaries and identifiers may change

Structure determines valid keys, comparisons, summaries, and integration choices. A file format alone does not describe that structure.

Metadata and provenance answer different questions

Source metadata

- definitions;
- methods;
- classifications;
- code lists; and
- quality flags.

Data dictionary

- variable names;
- meanings;
- types;
- units; and
- allowed values.

Provenance record

- provider and dataset;
- version and access date;
- license;
- retrieval method; and
- derived artifacts.

Metadata explain data; provenance explains their history.

Data quality means fitness for purpose

Completeness

Validity

Uniqueness

Consistency

Plausibility

Timeliness

- Quality requirements depend on the research question and intended use.
- Accuracy often cannot be established from a file alone.
- Provider flags and methodological notes are part of the evidence.
- A quality concern should be reported before it is corrected or excluded.

Application: validate before transforming

Structural checks

- expected columns and types;
- row count and temporal coverage;
- uniqueness of the proposed key;
- duplicate records; and
- allowed codes and units.

Semantic checks

- missingness patterns;
- impossible or suspicious values;
- consistency with definitions;
- agreement with metadata; and
- unresolved quality flags.

Validation should reveal assumptions and anomalies. It should not silently “fix” the evidence.

Organize the project by data role

```
analysis-project/  
|-- data-raw/  
|-- data-interim/  
|-- data-processed/  
|-- metadata/  
|-- scripts/  
'-- reports/
```

- Preserve raw inputs unchanged.
- Create derived data with code.
- Keep dictionaries, licenses and provenance near the project.
- Use stable names, not `final-v2-revised.csv`.
- Decide deliberately which data belong in Git.

Raw, interim and processed describe roles in a workflow—not levels of correctness.

Manage the project data responsibly

Before storing or sharing

- Check license and terms of use.
- Record citation and attribution.
- Confirm whether redistribution is permitted.
- Protect personal, confidential and location-sensitive data.

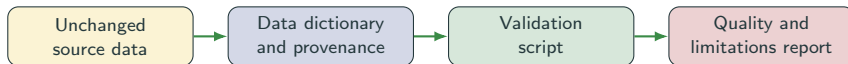
FAIR principles

- Findable
- Accessible
- Interoperable
- Reusable

FAIR does not necessarily mean open.

Farm and household data can create re-identification risks even after direct names are removed.

Key message: make data inspectable before analysis



1. State the grain and expected key.
2. Interpret variables, units, codes and flags.
3. Document source, meaning, license and history.
4. Validate structure and meaning without changing the raw file.
5. Record unresolved concerns for the next workflow stage.

The next session asks how to obtain and integrate multiple sources.

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